

Mystery Map: product requirements (MVP)

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Problem

Daily geography games test recall of names, shapes and positions, and a practised player runs out of things to learn from them. The player wants to reason about real facts of the world, with the map as the evidence.

Evidence so far comes from one player and four playable tests on 4 October 2026:

- A text round of clue-by-clue elimination was enjoyed in the moment, then judged a trivia quiz.
- Hand-written cases were rejected. Each one is played once and takes hours to write.
- A map of one hidden measure was played twice (median age, then hospital beds per 1,000 people) and loved both times.
- Comparison maps did not help identify the hidden measure. Hints about the layer itself did.

The product is a game in which the player works out what one unlabeled world map shows. Daily play is planned for later. The limit of the evidence is one player and two rounds of the final format.

Goals

The MVP succeeds when its maker plays nearly every map in it and nothing has to be rebuilt to go public.

Goal	How we will know
A round worth finishing	At least 90% of the available maps played within two weeks
Reasoning decides the score	Median score between 4 and 8 of 10 points
A deep supply of puzzles with no hand-written cases	300 validated layers, each added in 15 minutes or less of human time

Goal	How we will know
No ambiguity	Every layer and every option carries one exact definition, unit and data year
Ready to go public without rework	No MVP decision on data rights, borders, scheduling or saved state has to be reversed at launch

Non-goals

The MVP is one game mode on country-level data, with no server and no daily schedule.

Out of scope	Why
The other three modes (find the twin, cross-section, terrain route)	Untested. The engine must not block them, but none is built now
Accounts, a server, sync across devices	A static site is enough for this game and has nothing to maintain
Preventing cheating	Answers ship with the page, as in most daily games. Revisit only if leaderboards arrive
Leaderboards and multiplayer	They need accounts and a server
Technical layers such as ocean currents, and data below country level	Parked as a later advanced tier. The MVP uses everyday measures only
Comparison maps, pattern-breaker hints, yes-or-no questions	Tested or considered on 4 October and rejected
Making money	Not a goal of this version

User stories

There are two users: the player, and the curator who adds layers. At first both are the same person.

Player

- As a player, I want every finished map available to play in any order so that I can play as much as I like.
- As a player, I want to explore the map and see each country's name and group so that I can read the pattern myself.
- As a player, I want to switch between a globe and a flat map so that I can read the pattern in the view that suits me.
- As a player, I want to buy hints at different prices so that I can trade points for evidence when I am stuck.
- As a player, I want an exact definition on every option so that I never lose points to vague wording.
- As a player, I want the reveal to explain the pattern so that I learn why the map looks the way it does.
- As a player, I want my solved maps and past scores kept so that I can see how I am doing.
- As a player on a phone, I want every control to work by touch so that I can play anywhere.
- As a player, I want a round I left half-finished to be there when I return so that a reload costs me nothing.
- As a player, I want to be told when a country has no data, and not be charged for picking it.

Curator

- As the curator, I want a script that turns a public dataset into a validated layer so that adding a puzzle takes minutes.
- As the curator, I want the build to reject a layer with thin coverage, a missing definition or near-duplicate options so that a bad puzzle never ships.
- As the curator, I want a published day to keep its puzzle forever so that players who compare results saw the same map.

Requirements

Twelve requirements are must-haves. The MVP does not ship without all of them.

Must have (P0)

#	Requirement	Accepted when
1	Puzzle library: every validated layer is playable at any time, in any order	Unsolved maps are listed by number with no name. Solved maps show their name and score. Every layer keeps a permanent ID so a daily schedule can be added later
2	Map: countries coloured in five groups from less to more, grey for no data, with a legend, pan and zoom, and a switch above the map between a globe and a flat equal-area view	Every country with data can be selected at some zoom, small states included. Pointing at or tapping a country shows its name and group
3	Nothing leaks: no value, unit or layer name is visible before it is earned	Before the solve, values appear only for countries bought with the number hint
4	Extremes hint, 1 point	Names the three highest and three lowest countries, including any tied at third place
5	Number hint, 1 point per country	Shows the chosen country's real value with no unit. A country with no data, or one already bought, costs nothing
6	Unit hint, 3 points	States how the measure is counted, in one sentence
7	Guessing: six options, each with a one-line definition	The options are the answer plus five other layers from the catalog. None is a near-duplicate of the answer. A wrong guess costs 3 points and is struck out
8	Scoring: start at 10 points, never below 0	The score is fixed at the moment of the correct guess and shown at the reveal
9	Reveal: the layer's name, definition, unit, data year, source link, and a 2 to 4 sentence explanation	After the reveal, every country shows its real value
10	Saved state: the round in progress, solved maps and score history are kept in the browser	A reload mid-round restores points, hints bought and wrong guesses. Saved data carries a version number

#	Requirement	Accepted when
11	Catalog pipeline: a script builds the catalog and refuses bad layers	The build fails if a layer has data for under 80% of countries, lacks a definition, unit, year, source or licence, or has fewer than five eligible wrong options
12	Phone and access: playable by touch on a 360 px wide screen	Group is never shown by colour alone. All controls work by keyboard. The colour ramp stays readable with common colour blindness

Nice to have (P1)

Requirement	Accepted when
Share result	A copied text shows the map number, score and hints used, and nothing that reveals the answer
Daily schedule	One layer per calendar day, the same for every player. A date that has been published never changes its layer
Credits page	Every data source and the border data are named with their licences
Anonymous usage counts	Rounds started, solved, score and hints used are counted with no personal data and no cookies
First-visit walkthrough	A new player finishes a practice round before the first daily round
Report a problem	A player can flag a wrong figure or a border issue for a given day

Later (P2)

These are not built now, but the design must leave room for them: the other three modes, an advanced tier with technical layers, one-continent maps and other years to stretch the supply of puzzles, data below country level, accounts with sync, and answers held on a server.

Success metrics

Every target below is a starting hypothesis. There is no comparable game to take a benchmark from, so each one is reviewed after four weeks of play.

Private MVP, measured from the saved history in the browser

Metric	Target	What a miss means
Available maps played	90% within two weeks	The round is not worth coming back to
Median score	4 to 8 of 10	Above 8 the puzzles are too easy. Below 4 they are too hard or the hints are overpriced
Rounds that needed the unit hint	Under 30%	The cheaper hints are not giving enough to reason from
Median round length	2 to 6 minutes	Shorter means recognition, not reasoning. Longer means it will not fit a short break
Human time to add one layer	15 minutes or less	The catalog will not reach 300
Layers that pass validation but fail human review	Under 20%	The validation rules are too loose

After public launch, measured from anonymous usage counts

Metric	Target	Reviewed
Players who return within 7 days of a first round	25%	30 days after launch
Rounds started that are solved	80%	30 days after launch
Solved rounds that are shared	10%	30 days after launch

Decisions to get right from day one

Nine choices are cheap to make now and costly to change once people are playing. Each gets a fixed rule in the MVP.

Decision	Why it is costly to change later	Rule for the MVP
Puzzle schedule	Reshuffling changes maps people already played and compared	Each layer has a permanent ID from the start. The MVP has no schedule. When daily play is added, the date-to-layer list only ever grows at the end
Day boundary	Changing it gives some players two puzzles or none on the switch day	Deferred with daily play. Decide once before the first daily puzzle
Borders and country list	Every layer inherits them, and border complaints are the likeliest public criticism of a map game	Natural Earth shapes at 1:50 million scale. An entry exists for every ISO 3166 country code, which includes Palestine. Areas with no code are drawn but carry no data. Stated on the credits page
Data rights	A layer with unclear rights has to be pulled after launch	The catalog stores source, licence and year for every layer. Only data that allows reuse with credit is admitted
Projection	It changes how big each country looks, which changes how the puzzle reads	The player switches between a globe and a flat equal-area map with a control above the map
Game rules kept apart from the map	Rules tangled into map code cannot be tested, or reused by the next mode	Scoring, hints and option-picking are plain functions with their own tests, and know nothing about the map
Map as a swappable part	The later modes need rain and height layers the first map does not	The game talks to the map through a small fixed set of calls: colour the countries, report a tap, move the view

Decision	Why it is costly to change later	Rule for the MVP
Versioned data	An update that cannot read old saved rounds wipes streaks	The catalog file and the saved state each carry a version, with an upgrade step between versions
Checked explanations	One wrong fact at a reveal costs trust in all of them	Claude drafts every reveal explanation with its source. A person checks it before the layer goes live

Open questions

The three blocking questions were answered on 4 October, and nothing now blocks the technical design.

Question	Who answers	Status
Which projection? A street-map default stretches the far north, which distorts a map read by eye. The choices are that default, a globe, or a flat equal-area view drawn by deck.gl	Engineering	Answered: both views, with a switch
Is a tiled basemap needed under a map of coloured countries? If yes, which tile provider, and what are its terms and cost at public traffic?	Engineering	Answered: start with no tiled basemap
Which border dataset, and what is the policy for disputed areas and for territories that are not countries?	Product	Answered: Natural Earth shapes and the ISO country list
Does the day change at local midnight or at one global time?	Product	Deferred with daily play
Who writes the reveal explanations: drafted by an AI model and checked by a person, or written by hand?	Product	Answered: Claude drafts, a person checks

Question	Who answers	Status
Are the prices right? Hints at 1, 1 each and 3, and a wrong guess at 3, rest on two rounds of play	Product	Answered: keep these prices
Does the source catalog hold 300 everyday layers with data for 80% of countries? In plain terms: 300 is a guess that nobody has counted. Phase 0 counts the everyday measures that have figures for at least 4 in 5 countries	Data	Open. Counted in phase 0
Is the name Mystery Map free to use? It is a working title	Product	Open. One web search found no game with this exact name. Trademarks and the web address still need checking before launch

Phasing

There is no deadline. Each phase ends at a gate, and the next phase starts only when the gate is met.

1. **Phase 0, catalog.** Build the pipeline and the validation rules, and produce 30 layers.
 - Gate: 30 layers pass validation and human review, at 15 minutes or less each.
2. **Phase 1, private MVP.** Build all twelve must-haves and play on a private address.
 - Gate: 90% of the available maps played within two weeks, with a median score between 4 and 8.
3. **Phase 2, public launch.** Add the daily schedule, sharing, the credits page and usage counts, with at least 100 layers ready.
 - Gate: five outside players each finish a round with no help, and the border policy and data rights are checked.
4. **Phase 3, second mode.** Choose the next mode from the three parked ones and test it with a throwaway round first, as this mode was.